

Navigating the Commons: Online Appendices

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Appendix A. Technical appendix

A.1. Details of dynamic choice probabilities and moment conditions

I discretize the continuous capacity space into 53 grid levels, spaced at 150-ton increments, $\mathcal{K} = \{K^1 < K^2 < \dots < K^J\}$ where $K^1 = 150$, $K^J = 7,950$, and $J = 53$. I also discretize the productivity space into 11 grid levels, spaced logarithmically between the 1st and 99th percentiles of estimated productivity, with transition probabilities computed directly from the data. The whale stock enters the firm's state through its belief about next year's stock. I discretize the stock into 30 grid levels, spaced logarithmically between one third of the carrying capacity and the carrying capacity W_1 .

Following Hotz and Miller (1993) and Gowrisankaran and Schmidt-Dengler (2025), I define the choice-specific value for selecting capacity K^j as

$$(A.1) \quad v_{it}^j = -C(K^j, K_{it}; \gamma) + \rho(1 - p_t) \mathbb{E}_t [V_{t+1}(s_{it+1}, \psi_{t+1}; \Theta) | s_{it}, \psi_t]$$

where j indexes the j -th point of the capacity grid $\mathcal{K}$, and p_t is the wreck hazard.

From the incumbent's Bellman equation (12), the firm chooses K^j if and only if $v_{it}^j - I_{it}^j \epsilon_{it} \geq v_{it}^k - I_{it}^k \epsilon_{it}$, $\forall k \neq j$, where $I_{it}^j = K^j - (1 - \delta)K_{it}$. Monotonicity of $I(\cdot, K_{it})$ implies that for $1 \leq j < k \leq J$, the firm prefers K^j to K^k if and only if

$$\epsilon_{it} \geq \frac{v_{it}^k - v_{it}^j}{I_{it}^k - I_{it}^j} = \epsilon_{it}(j, k)$$

where the last equality defines the “ ϵ -cutoff” following Gowrisankaran and Schmidt-Dengler (2025). Intuitively, if the firm draws a higher (less favorable) cost shock than the cutoff, it chooses a smaller capacity. The firm chooses K^j if and only if $\underline{\epsilon}_{it}^j \leq \epsilon_{it} < \bar{\epsilon}_{it}^j$, where

$$\underline{\epsilon}_{it}^j = \begin{cases} \epsilon_{it}(j, j+1), & j < J, \\ -\infty, & j = J, \end{cases} \quad \bar{\epsilon}_{it}^j = \begin{cases} +\infty, & j = 1, \\ \epsilon_{it}(j-1, j), & j > 1. \end{cases}$$

Then the probability of choosing capacity K^j , conditional on staying ($a_{it} > 0$), is given by

$$(A.2) \quad \Pr_t(a_{it} = K^j | a_{it} > 0, s_{it}, \psi_t) = \Phi\left(\frac{\bar{\epsilon}_{it}^j}{\sigma}\right) - \Phi\left(\frac{\underline{\epsilon}_{it}^j}{\sigma}\right),$$

where $\Phi(\cdot)$ denotes the standard normal cumulative distribution function.

Another contribution of Gowrisankaran and Schmidt-Dengler (2025) is the computation of the conditional mean of the ϵ -shock without requiring numerical integration. By the properties of the truncated normal distribution, the conditional mean of the ϵ -shock is, $\forall j = 1, \dots, J$,

$$(A.3) \quad \mathbb{E}[\epsilon_{it} \mid a_{it} = K^j, s_{it}, \psi_t] = \sigma \cdot \frac{\phi(\underline{\epsilon}_{it}^j/\sigma) - \phi(\bar{\epsilon}_{it}^j/\sigma)}{\Phi(\bar{\epsilon}_{it}^j/\sigma) - \Phi(\underline{\epsilon}_{it}^j/\sigma)},$$

where $\phi(\cdot)$ is the standard normal probability density function. By using (A.1), (A.2), and (A.3), I define the expected continuation value as

$$(A.4) \quad V_{it}^{\text{cont}} = \sum_{j>0} \Pr_t(a_{it} = K^j \mid a_{it} > 0, s_{it}, \psi_t) \left\{ v_{it}^j - I_{it}^j \mathbb{E}[\epsilon_{it} \mid a_{it} = K^j, s_{it}, \psi_t] \right\}$$

From the expected continuation value and the logistic exit value draw $\zeta_{it} \sim \text{Logistic}(\bar{\zeta}, \sigma_L)$, the probability of exit is

$$(A.5) \quad \Pr_t(a_{it} = 0 \mid s_{it}, \psi_t) = \Pr(\zeta_{it} \geq V_{it}^{\text{cont}}) = \frac{1}{1 + \exp[(V_{it}^{\text{cont}} - \bar{\zeta})/\sigma_L]}.$$

Combining the period payoff (6), the exit probability (A.5), and the expected continuation value (A.4) yields a closed-form solution to the incumbent's Bellman equation:

$$V_t(s_{it}, \psi_t) = \Pi_t(s_{it}) + \mathbb{E}_\zeta[\max\{\zeta_{it}, V_{it}^{\text{cont}}\}] = \Pi_t(s_{it}) + \sigma_L \ln(e^{\bar{\zeta}/\sigma_L} + e^{V_{it}^{\text{cont}}/\sigma_L}).$$

The log-sum-exp term is the expected value of the better of exiting and staying under the logistic draw. The joint probability of staying and choosing capacity level K^j is given by

$$(A.6) \quad \Pr_t(a_{it} = K^j \mid s_{it}, \psi_t) = (1 - \Pr_t\{a_{it} = 0 \mid s_{it}, \psi_t\}) \Pr_t(a_{it} = K^j \mid a_{it} > 0, s_{it}, \psi_t).$$

Finally, given the logistic entry cost draw $\kappa_{it} \sim \text{Logistic}(\bar{\kappa}, \sigma_L)$, the probability of entry is

$$(A.7) \quad \Pr_t(a_{it}^{\text{pe}} = 1 \mid s_{it}, \psi_t) = \Pr(\kappa_{it} \leq \rho \mathbb{E}_t[V_{t+1}(s_{it+1}, \psi_{t+1}) \mid s_{it}, \psi_t]) \\ = \frac{1}{1 + \exp[-(\rho \mathbb{E}_t[V_{t+1}(s_{it+1}, \psi_{t+1}) \mid s_{it}, \psi_t] - \bar{\kappa})/\sigma_L]},$$

where a potential entrant's payoff-relevant state is $s_{it} = (x^e, W_t, K_t, Q_t)$, with x^e the common entry state.

Moment conditions. Equations (A.2), (A.5), and (A.7) give the model's counterpart of each decision in the data: the probability that a staying firm chooses each capacity level, the probability that an incumbent exits, and the probability that a potential entrant enters. Each depends on Θ through the equilibrium solve. A moment is the sample average of the gap between an observed decision and this model counterpart, either as it is or weighted by the firm's capacity, its productivity, or the year. The exit gap is the exit indicator minus the exit probability. The capacity gap is a function of the chosen capacity, such as whether the firm invested or how much, minus the model's expectation of that function over the firm's capacity choices. The entry gap is the number of entrants in a year minus the number the model expects from the pool of potential entrants.

The baseline set has 21 moments. Four match the exit margin: the exit rate and how it varies with capacity, productivity, and the year. Fifteen match the capacity margin among staying firms: whether the firm invests and whether it divests, each by capacity and productivity; how much it invests and how much it divests, each by capacity; the squares of the two amounts; whether it moves more than 300 tons up or down; and the dispersion of its net investment around the model's conditional mean. Two match the entry margin: the yearly count of entrants and its trend within the phase.

Identification comes from firms' observed decisions to enter, adjust capacity, and exit, given expected future profits across states and years within each phase. The level of exit identifies the mean exit value $\bar{\zeta}$, and the level of entry identifies the mean entry cost $\bar{\kappa}$. Given the means, the scale σ_L is identified by how much of exit and entry the state explains. A firm exits when its drawn exit value exceeds its continuation value, so σ_L sets how steeply exit responds to capacity, productivity, and the year, and how steeply entry responds to the year; a larger σ_L flattens these responses and leaves more exit and entry unexplained by the state. The frequencies of investing and divesting identify the fixed costs γ_0^+ and γ_0^- . Interacting the investment and divestment indicators with capacity separates the fixed costs from the per-ton terms. The mean amounts invested and divested identify the linear terms γ_1^+ and γ_1^- , and how the mean amounts respond to capacity, which moves the marginal value of a ton, identifies the convex terms γ_2^+ and γ_2^- , since a higher convex term damps that response. The dispersion of amounts among firms in the same state, in the squared amounts, their conditional variance, and the frequency of adjustments larger than 300 tons, identifies the standard deviation σ of the marginal adjustment cost shock jointly with the convex terms: a larger σ spreads the chosen levels around the level a firm would choose without the shock, while a larger convex term compresses that spread and thins the tail of large moves.

Estimator. I estimate Θ separately in each of six fifteen-year phases, 1815–29 through 1890–1905, by two-step GMM,

$$(A.8) \quad \hat{\Theta} = \underset{\Theta}{\operatorname{argmin}} \bar{g}(\Theta)' W \bar{g}(\Theta),$$

where $\bar{g}(\Theta)$ stacks the 21 sample moments and every evaluation re-solves the equilibrium of Section 3.4 at the trial Θ . The first step weights each moment by the inverse of its sample variance in the data. The second step uses the inverse of the covariance the moments would have if the decisions were drawn from the model's own probabilities at the first-step estimate. Standard errors follow the standard GMM formulas, with the covariance of the moments clustered by firm.

A.2. The planner's problem and the shadow value of a whale

This section derives the shadow value of a whale from the planner's problem of Section 5.1, and then describes the first-best computation.

The shadow value of a whale. The shadow value follows from the first-order condition of problem (18) in the year- $(t + 1)$ stock. Write $G_t := W_t + rW_t[1 - (W_t/W_1)^z] - Q_t$ for the stock that survives year t , so the constraint in (18) becomes $G_t \phi_{t+1}^W - W_{t+1}$. The stock W_{t+1} enters the objective in exactly three places,

$$\rho^t \Lambda_{t+1}^W (-W_{t+1}), \quad \rho^t W_{t+1}, \quad \rho^{t+1} \Lambda_{t+2}^W (G_{t+1} \phi_{t+2}^W - W_{t+2}),$$

the first because the constraint prices the stock the planner leaves for year $t + 1$, the second because that stock determines year- $(t + 1)$ harvest and hence welfare, and the third because it also determines what survives into year $t + 2$. Differentiating each and collecting,

$$-\rho^t \Lambda_{t+1}^W + \rho^t \frac{\partial W_{t+1}}{\partial W_{t+1}} + \rho^{t+1} \Lambda_{t+2}^W \phi_{t+2}^W \frac{\partial G_{t+1}}{\partial W_{t+1}} = 0,$$

and dividing by ρ^t gives

$$(A.9) \quad \Lambda_{t+1}^W = \frac{\partial W_{t+1}}{\partial W_{t+1}} + \rho \Lambda_{t+2}^W \frac{\partial W_{t+2}}{\partial W_{t+1}}, \quad \frac{\partial W_{t+2}}{\partial W_{t+1}} = \phi_{t+2}^W \frac{\partial G_{t+1}}{\partial W_{t+1}}, \quad \Lambda_{T+1}^W = 0.$$

The remaining derivative is the marginal welfare effect of an extra whale in year $t + 1$. The stock enters W_{t+1} only through industry harvest, $Q_{t+1} = \sum_{i \in \mathcal{I}_{t+1}} \mathcal{H}_{t+1}(x_{it+1}, W_{t+1}, K_{t+1})$,

so

$$\frac{\partial W_{t+1}}{\partial W_{t+1}} = \mathcal{P}_{t+1}(Q_{t+1}) \frac{\partial Q_{t+1}}{\partial W_{t+1}},$$

the price of a whale times the marginal effect of the whale stock on industry harvest, that is, the stock externality. Substituting it into (A.9) gives

$$\Lambda_{t+1}^W = \mathcal{P}_{t+1}(Q_{t+1}) \frac{\partial Q_{t+1}}{\partial W_{t+1}} + \rho \Lambda_{t+2}^W \frac{\partial W_{t+2}}{\partial W_{t+1}}.$$

Iterating forward gives the shadow value of a whale,

$$(A.10) \quad \Lambda_{t+1}^W = \sum_{\ell=t+1}^T \rho^{\ell-t-1} \left(\prod_{m=t+1}^{\ell-1} \frac{\partial W_{m+1}}{\partial W_m} \right) \mathcal{P}_\ell(Q_\ell) \frac{\partial Q_\ell}{\partial W_\ell},$$

the discounted value of every future harvest $\mathcal{P}_\ell \partial Q_\ell / \partial W_\ell$ that one more whale would have yielded, propagated forward through the stock dynamics $\partial W_{m+1} / \partial W_m$. Harvesting a whale in year t removes it from the year- $(t+1)$ stock, so its cost to society in year- t dollars is $\rho \Lambda_{t+1}^W$.

Computing the first best. The planner faces the same environment as the firms: the same cost and value draws $(\zeta_{it}, \epsilon_{it}, \kappa_{it})$, the same belief about next year's stock, and no information about the petroleum discovery before it arrives. The planner chooses rules that depend on the year and the firm's own state x_{it} , as the firms' decision rules do, and its value function has the same state s_{it} and the same conditional expectation $\mathbb{E}_t[\cdot | s_{it}, \psi_t]$ as the firms' problem (12). The planner uses one object a firm does not, the shadow value, the multiplier on the stock constraint.

The planner's flow payoff from firm i is the firm's profit (6) with two changes,

$$\left[\mathcal{P}_t(Q_t) - \rho \Lambda_{t+1}^W \right] \left(\mathcal{H}_t(x_{it}, W_t, K_t) + K_{it} \frac{\beta^K Q_t}{K_t} \right) - w_t^L L(K_{it}).$$

Each whale the firm harvests costs society $\rho \Lambda_{t+1}^W$ (the stock externality), so its harvest is valued at $\mathcal{P}_t - \rho \Lambda_{t+1}^W$. The firm's capacity also changes the other firms' harvest by $\beta^K Q_t / K_t$ whales per ton (the congestion externality).

The shadow value and the planner's rules determine each other. The shadow value (A.10) sums the future harvests one more whale would yield, and those harvests depend on the firms the planner's rules keep in the industry. The rules, in turn, price each whale at the shadow value. I therefore solve for the two together. Starting from candidate rules, I compute the industry and stock paths they imply and the shadow value along those paths.

I then re-optimize every rule against that shadow value and repeat until the rules converge.

Appendix B. Agents' supervision

The following examples are borrowed from Chapter 10 of Davis, Gallman, and Gleiter (2007) and Chapter 1 of Nicholas (2019).

A formal statement of the main outlines of the plan was usually given to the captain, as seen in the following passage from a letter dated 1 November 1834 from *Agent* Charles W. Morgan to *Captain* Reuben Russell, 2d.

The Bark being now ready for sea, as agent I have to advise you that she is bound on a whaling voyage to the Pacific Ocean-That she is fitted for thirty months-and that we wish you to cruise for sperm whales for 20 to 24 months and if not then full, fill up with whale Oil-we leave to your judgment the cruising ground on the Pacific though we would recommend the neighborhood of New Zealand, where both right & sperm whales are to be taken, and it would be well especially towards the end of the voyage to be where right whales could be taken. (Morgan Collection)

The following letter from agent Charles W. Morgan to a young captain, George H. Dexter, shows that agents also provided advice and encouragement.

As you have now taken the responsible station of Master of a Ship and are a young man you will permit me to offer some advice. The greatest difficulty I have observed with young Masters, is either too great indulgence or too great severity towards their crew. Discipline must be effectual, be administered with a steady hand especially among Sailors, and there is no station which requires more guard over the temper, than that of a master of a Ship. And on your first voyage depends in a great measure your future success in life. Let me then beg of you to keep a strict watch over the moral conduct of your crew, never permit your authority to be abused or set at naught, but at the same time never to use undue severity yourself or permit it in your officers. I have a real confidence in you and I trust you will not feel hurt at my giving advice which I feel it my duty to offer you. (Morgan Collection)

A glimpse of the exchanges between the captain and agent can be obtained from the letter books of whaling agents. For instance, on 21 November 1836, Charles W. Morgan wrote to George H. Dexter, a Morgan captain, at Montevideo, stating:

I wrote to you 15 Inst advising you that if you could get a price for your oil equal to 40 cts clear of every charge remitted home, you might sell it, but in consequence of news received of the failure of the Greenland Fishery, we think oil will be very high the next season. Therefore, we now advise that unless the oil will net 45 cts clear, we would wish you to bring it home. (Morgan Collection)

Similarly, in February 1858, Matthew Howland wrote to one of his captains, Philip Howland, advising him on his recent performance:

25 months out with 600 bbl sperm & 130 whale is rather low, but I am in hopes that you will come up now and be equal to any of them according to time out—I shall expect to hear of you into Talcuahana in March with from 800 to 1000 bbls of sperm oil on board. (Howland Collection)

Appendix C. Details of data

This section describes the data sources and modifications.

C.1. Voyage Database

Name of vessel.	Class.	Tonnage.	Captain.	Managing owner or agent.	Whaling-ground.	Date—		Result of voyage.			
						Of sailing.	Of arrival.	Sperm-oil.	Whale-oil.	Whalebone.	
1835.											
Nantucket, Mass.											
Barelay	Ship	30	Reuben Barney	Griffin Barney	Pacific Ocean	Nov. 13	—	J 1839	Bbls. 1,550	Bbls. 1,420	Lbs. 1,694
Baltic	do	410	William Keene	P. H. Folger	do	Sept. 8	Mar. 18, 1839		1,420	1,420	1,694
Columbus	do	314	Peter Coffin	Paul Mitchell's Sons	do	June 29	Nov. 12, 1838		1,398	1,398	16
Congress	do	339	William Upham	P. H. Folger	do	July 23	Nov. 20, 1838		1,902	1,902	
Catharine	do	324	Joseph M. Chase	Jared Coffin	do	July 29	Oct. 26, 1838		3,016	3,016	
Constitution	do	314	Edward C. Joy	C. G. & H. Coffin	do	Oct. 25	Apr. 7, 1839		1,630	1,630	
Eagle	do	335	Isaac Gardner	David Joy	Atlantic	July 29	Apr. 17, 1837		625	1,293	
Ganges	do	263	Barzilai T. Folger	Will am H. Gardner	Pacific Ocean	Oct. 26	May 10, 1839		1,644	1,644	
Harmony	Schooner	263	A. Swain	Thomas Coffin	Gulf of Mexico	Aug. 2	Aug. 20, 1836		260	156	
Howard	Ship	365	William Worth, 2d	S. & T. Hussey	Pacific Ocean	Sept. 21	Apr. 21, 1838		2,312	2,312	
John Adams	do	296	Obed Luce, Jr.	Griffin Barney	Atlantic & Ind	July 15	July 9, 1837		302	1,571	
Mary Mitchell	do	354	Samuel Joy	S. B. Tuck	Pacific Ocean	July 14	May 17, 1838		596	1,974	
Mary	do	36	Thomas Coffin, 2d	Daniel Jones	do	July 30	May 12, 1839		1,866	515	
Mount Vernon	do	344	Lewis B. Imbert	William Folger	do	Oct. 5	July 17, 1839		2,456	2,456	
President	do	293	Seth Cathcart	Joseph Starbuck	do	June 24	June 1, 1838		1,670	1,670	
Peru	do	257	William Brown, Jr.	David Joy	do	Oct. 4	Apr. 13, 1839		676	676	
Richard Mitchell	do	385	Henry C. Cleveland	P. Mitchell & Sons	do	July 20	Dec. 27, 1838		1,172	937	
Rambler	do	318	Robert M. McCleave	Aaron Mitchell	do	Sept. 8	Aug. 23, 1838		2,246	2,246	
Reaper	do	338	Timothy R. Coffin	P. H. Folger	do	Oct. 12					
Spartan	do	333	David W. Coffin	Daniel Jones	do	Oct. 4	May 4, 1839		1,790	1,790	

FIGURE C1. Sample of Starbuck (1878)

Source: Starbuck, A. 1878. *History of the American whale fishery from its earliest inception to the year 1876*.

The American Whaling Voyage database includes information about all known U.S. whaling voyages from the 1700s to the 1920s. Voyages have been defined based on customs house records, following Starbuck (1878), with each departure and subsequent return to

- 6 ABIGAIL, ship, of New Bedford. Registered (23) July 18, 1821 - permanent.
Built at Amesbury in 1810. 309 75/95 tons; length 97 ft., breadth 27 ft.,
depth 13 ft. 6 in. Master: Dennis Covell, Owners: Benjamin Rodman,
merchant, Andrew Robeson, David Coffin, Dennis Covell, Elisha Dunbar, New
Bedford. Two decks, three masts, square stern, no galleries, a bilbothead.
Previously registered at Newburyport Mar. 21, 1821.
- 7 Ship, of New Bedford. Re-registered (38) Nov. 17, 1823 - permanent.
Master: Hozekiah B. Gardner. Owners: Benjamin Rodman, Andrew Robeson,
David Coffin, Elisha Dunbar, merchants, New Bedford.
- 8 Ship, of New Bedford. Re-registered (47) Dec. 22, 1825 - permanent.
Master: Stephen Potter. Owners: Benjamin Rodman, merchant, Charles W.
Morgan, David Coffin, Elisha Dunbar, New Bedford.
- 9 Ship, of New Bedford. Re-registered (73) Nov. 19, 1831 - permanent.
Master: Benjamin Clark. Owners: Benjamin Rodman, Charles W. Morgan,
David Coffin, Elisha Dunbar, Benjamin Clark, New Bedford.
- 10 Ship, of New Bedford. Re-registered (43) June 13, 1835 - permanent.
Owners: Charles W. Morgan, William R. Rodman, David Coffin, Benjamin Clark,
Elisha Dunbar, New Bedford.

FIGURE C2. A sample of ship registers of New Bedford: Vessel *Abigail*

Source: Ship registers of New Bedford from HathiTrust (<https://www.hathitrust.org/>).

the port of origin constituting a single voyage. Figure C1 provides an example page from Starbuck (1878). A basic suite of information is included for most voyages, along with additional information on the ship's capacity and rig, declared destination, and amount of whale products. The database is digitized and provided by Mystic Seaport Museum and New Bedford Whaling Museum (<https://whalinghistory.org/>).

C.2. Recovering missing agents: vessel continuity and *Ship Registers*

The *Voyage Database* lists no whaling agent for 5,182 (33 percent) of the 15,707 voyages, and the gaps cluster before the 1830s. I recover as many of these agents as possible in two steps: vessel continuity and ship registers.

The first step exploits vessel continuity. A whaling agent is the one who outfits and manages a voyage and sells its catch; the agent runs the vessel, so it typically changes only when the vessel changes hands. When a voyage lists no agent but the same vessel has a named agent on a voyage within two years, I assign the missing voyage that agent. The vessel *Hannah and Eliza* of New Bedford, for example, lists Benjamin Rotch for its voyages of 1801 and 1804 and no agent for 1802 and 1803, and the rule assigns Rotch to both. This step recovers 1,113 voyages. Because it copies agents already in the database, it adds voyage-years to existing firms rather than creating new ones. To gauge the rule, I hold out the agent of each voyage that does record one and predict it from the same vessel's other voyages within two years. Of the 6,160 voyages with such a neighbor, the rule recovers the

recorded agent for 88 percent and the surname for 89 percent.

The second step reads the port *Ship Registers*, mainly New Bedford's, together with those of Boston, Salem, Providence, Newport, Bristol-Warren, Plymouth, and Newburyport. For a vessel with no named agent anywhere in the database, I locate the registration in force at the departure year, confirm the vessel by its tonnage, and take the first-named owner as the agent. Where the register records ownership shares or a named managing firm, I use the majority owner or that firm. The same reading also corrects entries the register contradicts, and I merge spellings and variants of one agent's name. These corrections are kept in a curated table that the cleaning step applies. It fills 975 further voyages, some run by firms otherwise absent from the database. Figure C2 shows the *Abigail* of New Bedford. The database lists Benjamin Rodman for 1825, no agent for 1829, and Charles W. Morgan from 1831. The register heads the owner list with Rodman at every registration from 1821 to 1831 and with Morgan from 1835, so the 1829 voyage is assigned to Rodman.

Together these steps raise agent coverage from 10,525 voyages (67 percent of the total) to 12,613, or 80 percent of all U.S. voyages.

C.3. Agency successions

The voyage register identifies a firm by the name of its managing agent. When a business passed to an heir, a partner, or a reorganized partnership, its vessels leave one name and reappear under another. Read literally, the register then shows a divestment by the old agent and an investment by the new one, and when the whole business moved at once, an exit and an entry. None of this is a decision to expand, contract, enter, or leave the industry. Left as recorded, it inflates measured turnover on all three margins.

I trace each vessel transfer between two named agents and decide whether it continues one firm or is a sale between two. The register contains 1,599 such transfers, 436,846 tons in all, forming 1,379 pairs of outgoing and incoming agent. Two kinds of evidence identify a succession. The first is scale. The outgoing agent hands over essentially its whole fleet, at least 90 percent of its tonnage in its last active year. Alternatively, the incoming agent begins with essentially the fleet it received, at least 90 percent of its first-year tonnage, in the year after the transfer. The second is a link between the two agents. Either the two names share a surname, or the transfer is too large for a routine sale, two or more vessels or at least 400 tons. When a pair passes a scale test and a link test, with the outgoing agent's business ending at the transfer, I merge the two. A pair whose incoming agent had already been operating for two years or more is merged only when the names are related. When an outgoing fleet is divided among several incoming agents, the firm's identity follows the largest share. Everything else stays as recorded. A single vessel passing between two agents

who both keep operating is a sale. The rules merge 165 pairs, 73,677 tons. The merged pairs include 21 chains of three or four successive names, each resolved to the last name in the chain.

C.4. Modifications

For the purpose of empirical analysis, the output quantities (in barrels of sperm oil and whale oil, and pounds of whalebone) are converted into the number of whales harvested. Following Scammon (1874), this paper assumes that an average sperm whale taken yielded 25 barrels of sperm oil and an average baleen whale taken yielded 60 barrels of whale oil. Historical accounts also document that around 10% of whales escaped but subsequently died from their wounds. Therefore, I relate the barrels of oil observed in the data to the number of whales harvested in the following way:

$$Q_{it}^{SW} = \frac{Q_{it}^{soil}}{25} + 0.1 \times \frac{Q_{it}^{soil}}{25}$$

$$Q_{it}^{BWO} = \frac{Q_{it}^{woil}}{60} + 0.1 \times \frac{Q_{it}^{woil}}{60}$$

where Q_{it}^{SW} is the number of sperm whales harvested by firm i in year t , Q_{it}^{soil} is the barrels of sperm oil, Q_{it}^{BWO} is the number of baleen whales harvested to make whale oil, and Q_{it}^{woil} is the barrels of whale oil. From the logbook database,¹ I assume that an average baleen whale taken yielded 500 pounds of whalebone.

$$Q_{it}^{BWB} = \frac{Q_{it}^{bone}}{500} + 0.1 \times \frac{Q_{it}^{bone}}{500}$$

where Q_{it}^{BWB} is the number of baleen whales harvested to make whalebone and Q_{it}^{bone} is the pounds of whalebone. Then, the number of baleen whales harvested, Q_{it}^{BW} , is determined as follows:

$$Q_{it}^{BW} = \max \{ Q_{it}^{BWO}, Q_{it}^{BWB} \}$$

¹The American Offshore Whaling Log database comprises information from 1,381 logbooks documenting U.S. whaling voyages spanning 1784 to 1920. This dataset was extracted from the original whaling logbooks during three distinct scientific research projects. The first initiative was led by Lt. Cmdr. Matthew Fontaine Maury in the 1850s, the second by Charles Haskins Townsend in the 1930s, and the third by a team from the Census of Marine Life project (CoML, www.coml.org), led by Tim Denis Smith from 2000 to 2010. The CoML team assembled the Maury and Townsend data from archival sources. The data file encompasses 466,134 records organized in a standardized format, including voyage ID, coordinates (latitude/longitude), date, whale encounter, species, harpooned, place, and so on.

Finally, the whaling output by firm i in year t is defined as the total number of whales harvested:

$$Q_{it} = Q_{it}^{SW} + Q_{it}^{BW}$$

Appendix D. Estimation strategy for a nonstationary global whaling market

I use a full-solution method instead of conditional choice probability approaches (e.g., Bajari, Benkard, and Levin 2007) for two reasons.

The first reason relates to counterfactual analysis. This paper evaluates firms' behavioral responses to regulatory, environmental, and economic changes in exploiting shared resources. Such analysis requires counterfactual equilibrium decision rules obtained by re-solving the model under alternative conditions. For example, imposing a tax on whaling output raises costs and alters firms' equilibrium behavior. To assess these changes, the model must be solved again. A full-solution approach ensures consistency between estimation and counterfactual exercises, enabling direct and reliable comparisons of model outcomes.

The second reason relates to estimation. The global whaling industry presents significant empirical challenges due to the nonstationary evolution of economic fundamentals.² Figures 1, 2, and E1 illustrate the dynamics of American whaling, including demand shifts, whale stock transition, and technological advancements. Since this paper examines whaling firms' entry, exit, and investment decisions in such a nonstationary environment, the firms' policy functions must adapt over time. In essence, time becomes an additional state variable, making policy functions explicitly time-dependent, as detailed in Section 3.3.

The cross-sectional and time-series characteristics of the data reduce the effective sample size for every year, complicating the application of two-step conditional choice probability methods (e.g., Aguirregabiria and Mira 2007; Bajari, Benkard, and Levin 2007; Pakes, Ostrovsky, and Berry 2007; Pesendorfer and Schmidt-Dengler 2008). First-step nonparametric estimates of conditional choice probabilities become imprecise when only a few or no choices are observed in certain states. The model has to fill in these blanks, and hence this paper takes a full-solution approach instead, as in Benkard (2004), Goettler and Gordon (2011), Igami (2017), and Elliott (2024).

²Nonstationarity is particularly important in resource industries. For example, the American whaling industry experienced rapid growth and decline over a century due to changes in whale stocks, alternative resource availability, and technological advancements. Similar trends can be observed in modern energy sectors, such as cost reductions and increasing demand for solar panels, electric vehicles, and other renewables.

Appendix E. Additional tables and figures

TABLE E1. Demand-side statistics for the American whaling industry

Variable	Unit	Mean	SD	Min	Max
Whales demanded	Number	4083.60	3776.81	23.21	11689.03
Whale price	1880 dollars per whale	798.20	325.06	296.84	2013.65
US real GDP per capita	1996 dollars	2434.03	1103.68	1249.00	5357.00
Petroleum price	1880 dollars	0.05	0.06	0.01	0.41
Year		1856.50	30.74	1804.00	1909.00

Notes: The dataset comprises 106 observations. One exception is petroleum price data, which begins in 1859, the year of the petroleum discovery.

TABLE E2. Dynamic cost estimates under alternative stock-belief uncertainty σ_W

	Unit	0.05	0.10	0.20
Mean of entry costs: $\bar{\kappa}$	\$000	614.10 (43.87)	892.66 (70.12)	1036.53 (85.18)
Mean of exit scrap values: $\bar{\zeta}$	\$000	411.96 (31.04)	655.99 (55.86)	772.27 (67.82)
SD of marginal adj. costs: σ	\$/ton	206.38 (25.46)	227.45 (30.16)	272.36 (37.28)
Investment fixed: γ_0^+	\$000	3.83 (1.09)	5.04 (1.26)	6.30 (1.53)
Investment linear: γ_1^+	\$/ton	365.99 (25.33)	363.56 (27.88)	391.18 (32.33)
Investment convex: γ_2^+	\$/ton ²	0.26 (0.03)	0.28 (0.04)	0.33 (0.05)
Divestment fixed: γ_0^-	\$000	35.71 (4.36)	40.04 (5.50)	47.53 (7.06)
Divestment linear: γ_1^-	\$/ton	113.72 (25.16)	111.50 (26.32)	94.46 (30.72)
Divestment convex: γ_2^-	\$/ton ²	0.23 (0.03)	0.26 (0.04)	0.32 (0.05)

Note: σ_W is the standard deviation, in log points, of firms' beliefs about next year's whale stock; the baseline sets $\sigma_W = 0.10$ (Section 4.2). Each column re-estimates the dynamic cost parameters holding σ_W fixed at the stated value. Estimates are for the 1845–59 phase; standard errors are in parentheses. Baseline estimates are in Table 5.

TABLE E3. Sensitivity to the post-1859 demand elasticity

	Post-1859 price elasticity α_p		
	-2.0	-4.39 (baseline)	-5.0
Laissez-faire (\$M)	32.19	30.32	30.14
First best (\$M)	47.42	44.68	44.48
Loss (\$M)	15.23	14.36	14.34
Share of first best	32.1%	32.1%	32.2%

Note: The price elasticity α_p is estimated on the pre-petroleum sample and imposed on the post-1859 period. This table re-estimates the post-1859 demand shifters and re-solves the laissez-faire and first-best allocations at $\alpha_p = -2.0$ and -5.0 , holding all other primitives at their baseline values. Welfare is in millions of 1880 dollars, discounted to 1815.

TABLE E4. Sensitivity to the production-function capacity elasticity β_k

		Baseline	$\hat{\beta}_k + 2SE$	$\hat{\beta}_k + 4SE$
Panel A. Production function				
Own capacity β_k		1.048	1.063	1.079
Productivity persistence ρ		0.300	0.645	0.749
Congestion β_K		-0.057	-0.060	-0.062
Whale stock β_w		5.101	5.349	5.545
Time trend β_t		0.0135	0.0137	0.0138
Panel B. Dynamic cost estimates (phase: 1845–59)				
Mean of entry costs: $\bar{\kappa}$	\$000	892.66 (70.12)	887.59 (71.83)	868.35 (72.18)
Mean of exit scrap values: $\bar{\zeta}$	\$000	655.99 (55.86)	647.65 (57.01)	631.85 (56.78)
SD of marginal adj. costs: σ	\$/ton	227.45 (30.16)	237.47 (32.16)	254.20 (34.74)
Investment fixed: γ_0^+	\$000	5.04 (1.26)	5.32 (1.33)	5.74 (1.43)
Investment linear: γ_1^+	\$/ton	363.56 (27.88)	369.32 (29.04)	380.06 (30.56)
Investment convex: γ_2^+	\$/ton ²	0.28 (0.04)	0.30 (0.04)	0.32 (0.04)
Divestment fixed: γ_0^-	\$000	40.04 (5.50)	41.79 (5.82)	44.69 (6.37)
Divestment linear: γ_1^-	\$/ton	111.50 (26.32)	107.92 (27.89)	102.58 (30.62)
Divestment convex: γ_2^-	\$/ton ²	0.26 (0.04)	0.28 (0.04)	0.30 (0.05)

Note: This table raises $\hat{\beta}_k$ by two and four first-stage standard errors and re-estimates the production function (Panel A) and the dynamic cost parameters (Panel B). Panel B reports the 1845–59 phase; standard errors are in parentheses. Baseline estimates are in Table 5.

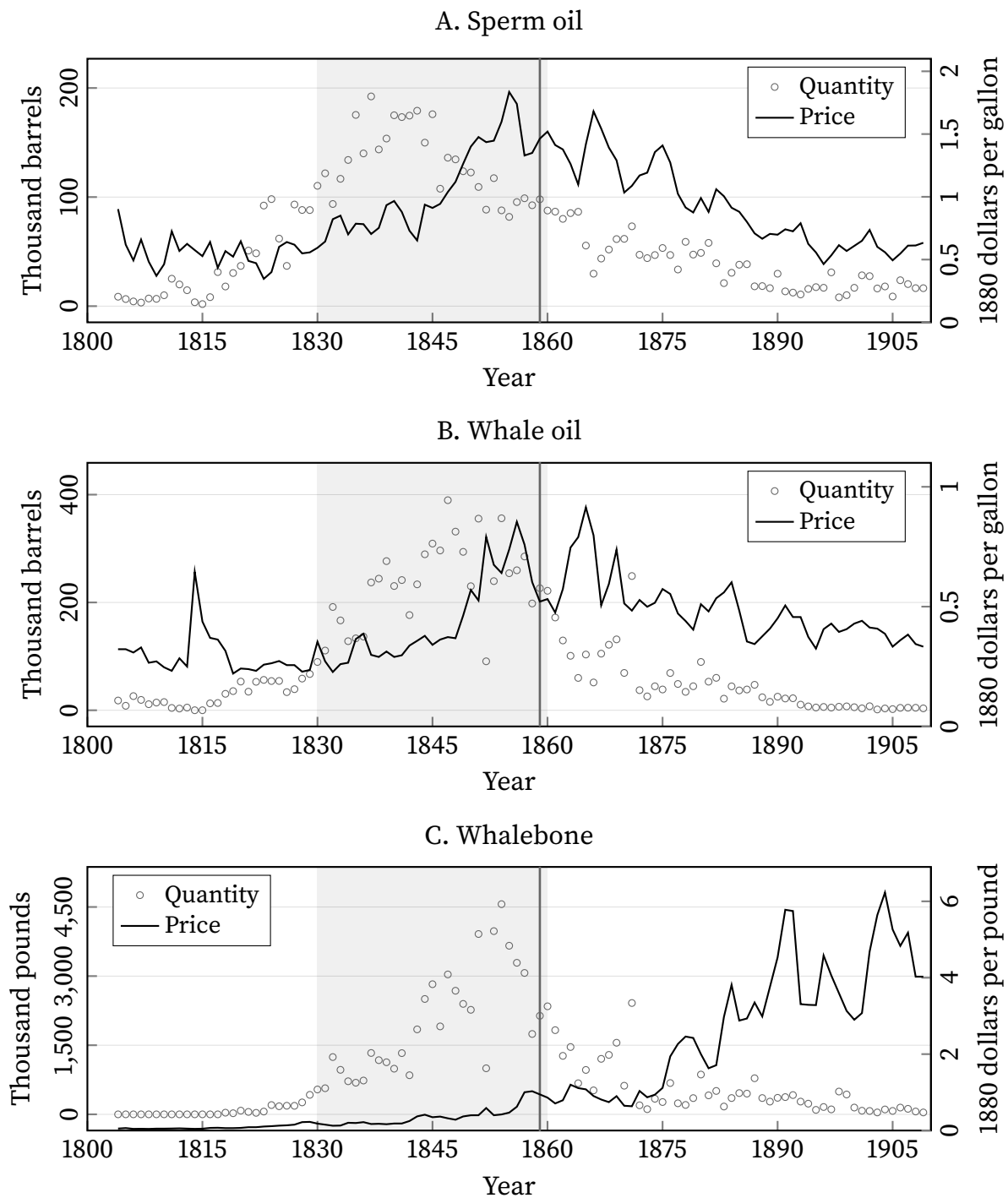


FIGURE E1. Quantities and prices for whaling products

Note: The term “Golden Age” refers to the period when the American whaling industry was at its peak, from the 1830s to the 1850s. The year of petroleum discovery in 1859 is indicated by the vertical red line.

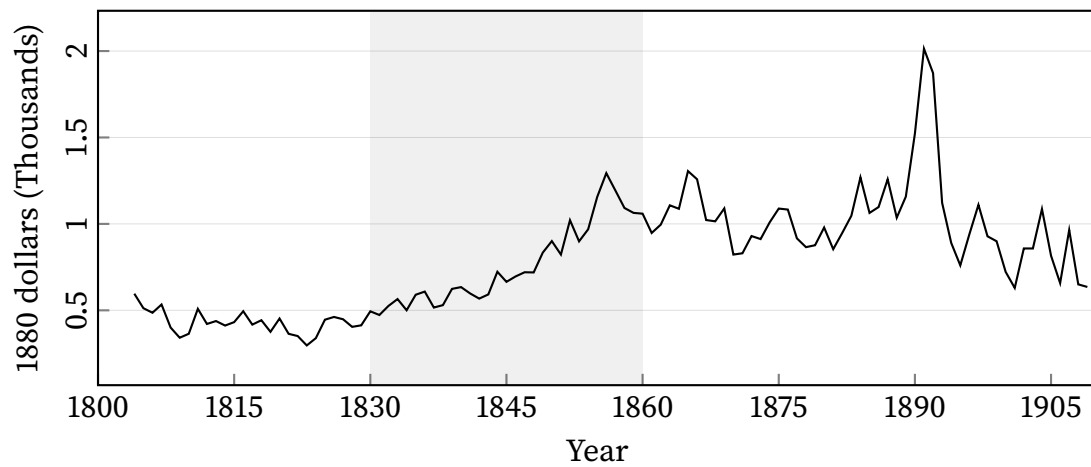


FIGURE E2. Whale price

Note: The prices of whaling products come from Tower (1907) and Davis, Gallman, and Gleiter (2007). I calculate the price *per whale* as total output value divided by total whales harvested each year. The output value equals sperm oil quantity $\times$ price + whale oil quantity $\times$ price + whalebone quantity $\times$ price. I compute total whales harvested by summing catches across all voyages.

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